

Panduan API integrasi — SKU No

Dokumen ini untuk **sistem eksternal** (ERP, middleware, batch job) yang memanggil pipeline DDMRP per SKU tanpa melalui UI.

Aplikasi (UI): <https://transtrack-ddmrp.skom.my.id>

Base URL API (produksi): <https://transtrack-ddmrp-api.skom.my.id>

Prefix: /api/analytics

Lingkungan	Base URL API
Produksi	https://transtrack-ddmrp-api.skom.my.id
Development (Docker lokal)	http://localhost:8000

Semua endpoint integrasi memakai parameter **sku_no** (setara **sku** di API internal).

Ringkasan endpoint

Method	Path	Fungsi
POST	/api/analytics/integration/run	Function run optimize + simpan buffer & latest run
GET	/api/analytics/integration/result?sku_no=<SKU_NO>	Detail on result (kasus optimize)
GET	/api/analytics/integration/replenishment?sku_no=<SKU_NO>	Detail on replenishment + snapshot operasional

Prasyarat: SKU harus ada di **sku_master** dan punya data demand (daily_record) sebelum **integration/run**.

1. POST /api/analytics/integration/run

Menjalankan pipeline penuh (sama dengan POST /api/analytics/run) dengan body berbasis sku_no.

Request

Content-Type: application/json

Field	Wajib	Default	Keterangan
sku_no	ya	—	Kode SKU (string, trim)
sl_target	tidak	0.95	Target service level GA (0.5–0.999)
pop_size	tidak	30	Ukuran populasi GA (6–80)
n_gen	tidak	80	Generasi GA (5–120)
include_baseline	tidak	true	Sertakan perbandingan baseline

```
{
  "sku_no": "1000001",
  "sl_target": 0.95,
  "pop_size": 30,
  "n_gen": 80,
  "include_baseline": true
}
```

Response 200

```
{
  "sku_no": "1000001",
  "status": "ok",
  "buffer_id": 123,
  "latest_run_id": 77,
  "forecast": { "...": "..." },
  "optimize": { "...": "..." }
}
```

- `buffer_id` — buffer **Active** yang disimpan.
- `forecast` — model terbaik, metrik, ADU, unit.
- `optimize` — VF/LTF optimal, KPI, TOR/TOY/TOG (di dalam `optimized.kpi`).

Error umum

HTTP	Penyebab
404	SKU tidak ada atau tidak ada data demand
500	Kegagalan forecast/GA (detail di body)

Contoh cURL

```
curl -s -X POST "https://transtrack-ddmrp-api.skom.my.id/api/analytics/integration/run" \
  -H "Content-Type: application/json" \
  -d '{"sku_no":"1000001","sl_target":0.95,"pop_size":30,"n_gen":80}'
```

2. GET /api/analytics/integration/result?sku_no=<SKU_NO>

Mengambil **latest run** tersimpan untuk SKU (tanpa menjalankan ulang pipeline).

Query

Parameter	Wajib	Keterangan
sku_no	ya	Kode SKU

Response 200 (ada run)

```
{
  "sku_no": "1000001",
  "status": "ok",
  "sku": "1000001",
  "latest_run": {
    "id": 77,
    "run_at": "2026-04-08T18:12:34.123456",
    "unit": "PCS",
    "forecast": { "sku": "1000001", "best_model": "...", "adu": 15.43 },
    "optimize": { "sku": "1000001", "optimized": { "fv_opt": 0.1, "ltf_opt": 1.0, "kpi": {} } }
  }
}
```

Response 200 (belum pernah run)

```
{
  "sku_no": "1000001",
  "status": "ok",
  "sku": "1000001",
  "latest_run": null
}
```

Error

HTTP	Penyebab
400	Parameter sku_no kosong

```
curl -s "https://transtrack-ddmrp-api.skom.my.id/api/analytics/integration/result?sku_no=1000001"
```

3. GET /api/analytics/integration/replenishment?sku_no=<SKU_NO>

Mengambil rencana replenishment dari **buffer aktif** terakhir, termasuk posisi operasional hari ini.

Query

Parameter	Wajib	Keterangan
sku_no	ya	Kode SKU

Response 200

```
{
  "sku_no": "1000001",
  "status": "ok",
  "sku": "1000001",
  "unit": "PCS",
  "buffer_id": 123,
  "version": 1,
  "today_date": "2026-04-09",
  "end_date": "2026-04-15",
  "leadtime_days": 7,
  "adu": 15.43,
  "vf_opt": 0.1,
  "ltf_opt": 1.0,
  "tor": 58.0,
  "toy": 196.0,
  "tog": 340.0,
  "operational": {
    "on_hand": 196.0,
    "open_order": 0.0,
    "qualified_demand": 15.4,
    "nfe": 58.45,
    "zone": "YELLOW",
    "suggested_order_qty": 171.0,
    "confirmed_pos": [],
    "unit": "PCS"
  },
  "recommendations": [
    {
      "date": "2026-04-09",
      "order_qty": 12.0,
      "nfe": 3.5,
      "zone": "RED"
    }
  ]
}
```

```
    ]
}
```

Field penting

Field	Arti
tor, toy, tog	Batas zona buffer (dari RUN 3)
operational.nfe	NFE operasional hari <code>today_date</code>
operational.suggested_order_qty	Saran qty order setelah OP/QD terkini
operational.confirmed_pos	PO confirmed yang belum diterima
recommendations[].order_qty	Saran residual per hari di jendela buffer

Error

HTTP	Penyebab
404	Tidak ada buffer Active — jalankan <code>integration/run</code> dulu
400	<code>sku_no</code> kosong

```
curl -s "https://transtrack-ddmrp-api.skom.my.id/api/analytics/integration/replenishment?sku="
```

4. Alur integrasi yang disarankan

```
sequenceDiagram
```

```

    participant ERP
    participant API as IDAS API
    ERP->>API: POST /integration/run (sku_no)
    API-->>ERP: buffer_id, forecast, optimize
    ERP->>API: GET /integration/replenishment?sku_no=
    API-->>ERP: operational + recommendations
    Note over ERP: Buat PO di ERP atau via /api/purchase-orders
    ERP->>API: GET /integration/result?sku_no= (audit / sync)

```

1. Pastikan master + demand sudah di-load (UI upload atau impor CLI).
2. POST `/integration/run` untuk SKU yang perlu diperbarui.
3. GET `/integration/replenishment` untuk keputusan order harian.
4. Opsional: konfirmasi PO lewat POST `/api/purchase-orders` lalu panggil lagi replenishment atau POST `/api/analytics/recalc-operational?sku=`.

5. Endpoint terkait (bukan prefix integrasi)

Endpoint	Keterangan
GET /api/analytics/dataset-status	Cek kesiapan data
GET /api/analytics/latest-run?sku=	Sama seperti result, parameter sku
GET /api/analytics/replenishment?sku=	Sama seperti integration replenishment
POST /api/purchase-orders	PO operasional
POST /api/analytics/recalc-operational?sku=	Recalc NFE tanpa GA

6. Perbedaan parameter sku vs sku_no

Konteks	Parameter
UI / API internal	sku (query atau body RunBody)
API integrasi	sku_no (query atau body IntegrationRunBody)

Nilai dan normalisasi (trim string) sama; hanya nama field yang berbeda untuk kontrak integrasi.

Dokumen terkait

- USER_MANUAL.md — panduan UI
- INSTALLATION.md — instalasi & reset DB
- ../README.md — PO API, scheduler, dashboard